

Tutorial No : 4

Tutorial Title: Write a program in C++ to perform arithmetic operations on complex numbers. Design the class for complex number representation and the operations to be performed. The objective of this assignment is to learn the concepts of classes and objects..

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Program1:

```
#include <iostream>
using namespace std;

class complex {
    int real, img;
public:
    void accept(int r, int i) {
        real = r;
        img = i;
    }
    void display() {
        cout << "\nThe complex number is:";
        cout << real << "+i" << img;
    }
};

int main() {
    complex c1, c2;
    c1.accept(3, 5);
    c2.accept(9, 20);
    c1.display();
    c2.display();
    return 0;
}
```

Output:

```
The complex number is:3+i5
The complex number is:9+i20

=== Code Execution Successful ===
```

Program 2:

```
#include <iostream>
using namespace std;

class complex {
    float real, img;
public:
    void accept(float r, float i) {
        real = r;
        img = i;
    }

    void display() {
        if (img >= 0) {
            cout << real << " + " << img << "i";
        } else {
            cout << real << " - " << -img << "i";
        }
    }

    void add(complex c1, complex c2) {
        real = c1.real + c2.real;
        img = c1.img + c2.img;
    }

    void sub(complex c1, complex c2) {
        real = c1.real - c2.real;
        img = c1.img - c2.img;
    }

    void mult(complex c1, complex c2) {
        real = (c1.real * c2.real) - (c1.img * c2.img);
        img = (c1.real * c2.img) + (c1.img * c2.real);
    }

    void divide(complex c1, complex c2) {
        float denominator = (c2.real * c2.real) + (c2.img * c2.img);
        real = (c1.real * c2.real + c1.img * c2.img) / denominator;
        img = (c1.img * c2.real - c1.real * c2.img) / denominator;
    }
};

int main() {
```

```
complex c1, c2, c3, c4, c5, c6;
c1.accept(3, -5);
c2.accept(9, 20);

cout << "c1 = ";
c1.display();
cout << "\nc2 = ";
c2.display();

c3.add(c1, c2);
cout << "\n\nAddition: ";
c3.display();

c4.sub(c1, c2);
cout << "\n\nSubtraction: ";
c4.display();

c5.mult(c1, c2);
cout << "\n\nMultiplication: ";
c5.display();

c6.divide(c1, c2);
cout << "\n\nDivision: ";
c6.display();
cout << endl;

return 0;
}
```

Output:

```
c1 = 3 - 5i
c2 = 9 + 20i

Addition: 12 + 15i
Subtraction: -6 - 25i
Multiplication: 127 + 15i
Division: -0.151767 - 0.218295i
```

